

CESAR ARGUELLO

cesar.n.arguello.martinez.gr@dartmouth.edu $\diamond$ carguellom.com | github.com/carguelloM

EDUCATION

Ph.D. in Computer Science

Dartmouth College

September 2022 – Present

Hanover, NH, USA

Advisor: Dr. David Kotz

Research area: Security and privacy in the lifecycle of IoT for consumer environments

B.S. in Physics & Computer Science

University of Florida

August 2018 – May 2022

Gainesville, FL, USA

International Baccalaureate Diploma Programme

UWC Red Cross Nordic

August 2016 – May 2018

Flekke, Norway

PUBLICATIONS

Conference Papers

- CP1** Ravindra Mangar, **Cesar Arguello**, David Inyangson, Tina Pavlovich, Karen Gareis, and Tushar Jois. Engaging students from under-represented groups to pursue graduate school in computer science and engineering. *Proceedings of the 56th ACM Technical Symposium on Computer Science Education (SIGCSE TS)*, 2025. Acceptance rate 33%.
- CP2** **Cesar Arguello**, Beatrice Perez, Timothy J. Pierson, and David Kotz. Detecting battery cells with harmonic radar. *Proceedings of the ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2024. Acceptance rate 21%.
- CP3** Beatrice Perez, **Cesar Arguello**, Timothy J. Pierson, Gregory Mazzaro, and David Kotz. Evaluating the practical range of harmonic radar to detect smart electronics. *Proceedings of the IEEE Military Communications Conference (MILCOM)*, 2023. Acceptance rate 40%.

Journal Articles

- JA1** Timothy J. Pierson, **Cesar Arguello**, Beatrice Perez, Wondimu Zegeye, Kevin Kornegay, Carl Gunter, and David Kotz. We need a “building inspector for IoT” when smart homes are sold. *IEEE Security & Privacy*, 2024.

Posters

- P1** **Cesar Arguello**, Hunter Searle, Sara Rampazzi, and Kevin Butler. [Poster] A practical methodology for ML-based EM side-channel disassemblers. *Proceedings of the Poster Session of the 7th IEEE European Symposium on Security and Privacy (EuroS&P)*, 2022.

PATENTS

- **Cesar Arguello**, Beatrice Perez, Timothy J. Pierson, and David Kotz. Harmonic radar for battery detection. U.S. Patent Application US19/216,352. Filed May 22, 2025. (*Pending.*)

RESEARCH EXPERIENCE

Graduate Research Assistant

Dartmouth College

Department of Computer Science

Advisor: Dr. David Kotz and Dr. Timothy Pierson

January 2023 – Present

Hanover, NH, USA

- Design, implement, and debug embedded systems across the hardware–software stack in C, C++, Python, and Verilog, building real-time wireless systems on software-defined radio platforms.
- Design and implement experiments and circuit prototypes for novel applications of harmonic radar in IoT security, spanning theoretical modeling through field measurement.
- Develop research plans targeting conference and journal publication, and review and synthesize the IoT security literature to position new work.

Research Assistant

University of Florida

Florida Institute for Cybersecurity Research

Advisor: Dr. Sara Rampazzi and Dr. Kevin Butler

June 2021 – May 2022

Gainesville, FL, USA

- Designed and ran experiments to test theoretical frameworks for EM side-channel disassembly.
- Collected, processed, and analyzed EM traces using statistical and machine learning models.
- Developed research plans leading to project deliverables, including a peer-reviewed poster.

Research Assistant

University of Florida

Department of Physics, SuperCDMS HVeV-DMC Data Analysis and Simulations Team

Advisor: Dr. Tarek Saab

May 2021 – May 2022

Gainesville, FL, USA

- Built Ionization Measurement with Phonons At Cryogenic Temperatures (IMPACT) simulation files for the Geant4 simulator.
- Analyzed Geant4 simulation output in Python.
- Led group discussions on simulation results.

Research Assistant

University of Florida

Department of Mechanical and Aerospace Engineering, ADAMUS Lab

Advisor: Dr. Riccardo Bevilacqua

August 2020 – February 2021

Gainesville, FL, USA

- Designed, developed, and deployed C++ libraries controlling a CubeSat's magnetometers, EPS, and battery over I²C, integrated with a BeagleBone Black microcontroller.
- Tested and debugged flight component prototypes.
- Authored weekly reports on flight software progress, blockers, and design decisions.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Dartmouth College

CS 50: Software Design and Implementation

September 2022 – January 2023

Hanover, NH, USA

- Built and maintained automated grading scripts in Bash and Python for student assignments.
- Conducted code reviews for programming projects, giving feedback on correctness, efficiency, and style.
- Held office hours providing one-on-one mentorship on course material and projects.

Undergraduate Teaching Assistant
University of Florida
CDA 3101: Introduction to Computer Organization

January 2021 – May 2021
Gainesville, FL, USA

- Led discussion sections on lecture material to help students meet course learning objectives.
- Managed assignment evaluation and provided detailed written feedback.
- Held office hours to review course material and answer questions on assignments.

INDUSTRY EXPERIENCE

Product Development Engineer Intern
Intel Corporation

January 2022 – May 2022
Santa Clara, CA, USA

- Designed, developed, and debugged sort test programs for server products.
- Tested, validated, modified, and redesigned circuits to guarantee component margin to specification.
- Evaluated component specification against measured performance to match component requirements with production equipment capability.

SERVICE

Artifact Evaluation Committee, Network and Distributed System Security Symposium (NDSS) 2027
External Reviewer, IEEE GLOBECOM 2026

HONORS & AWARDS

Cum Laude, B.S., University of Florida
Phi Beta Kappa, Honor Society
Davis United World College Scholar

May 2022
May 2022
August 2018

TECHNICAL SKILLS

Programming	Python, C, C++, Verilog, MATLAB, Bash
Wireless & SDR	GNU Radio, USRP, HackRF, ADALM-Pluto/AntSDR, gr-ieee802-11, openwifi, 802.11 PHY/MAC
Security tooling	Wireshark, tcpdump, Scapy, aircrack-ng, hostapd, wpa_supplicant
Libraries	PyTorch, NumPy, SciPy, scikit-learn, Matplotlib
Tools	Git, GDB, Vivado, Docker, Linux, L ^A T _E X
Spoken languages	English, Spanish